Xuanpei Chen

Manchester | caesar20040723@gmail.com | caesar723.github.io/myCV | github.com/Caesar723

EDUCATION

The University of Manchester, BSc in Computer Science — Sep. 2024 - Dec. 2027

Areas of Interest: Reinforcement Learning, Backend Development, Game Development

Shanghai Thomas School — Jul. 2021 - Jul. 2023

A-levels: Mathematics(A*), Further Mathematics(A*), Physics(A*), Computer Science(A)

EXPERIENCE

MuteGravity-Face Swap AI
Jun. 18, 2025 - Aug. 20, 2025

Internship: AI Programmer — Shanghai, China

- Independently led the development of a complete facial expression parameter conversion project, transforming real human face videos into MetaHuman parameters; the project was successfully adopted and applied within the company.
- Utilized Pixel3DMM to generate normal maps for lighting augmentation, and wrote/optimized Unreal Engine scripts to collect high-precision MetaHuman expression images, improving efficiency from 2s per image to 50 images per second.
- Built a lightweight model based on VAE-GAN to achieve high-quality conversion between MetaHuman and real human face data, expanding training datasets and improving final model performance.
- Employed DINOv3 for feature extraction and combined with a temporal Transformer to generate stable, jitter-free facial parameters, greatly enhancing the generalization and robustness of video-to-parameter conversion.

Iluvatar CoreX
Jul. 2024 - Sep. 2024

Internship: Test Engineering Engineer — Shanghai, China

- Independently developed the "Auto Dump" project, automating model operator dumping with Jenkins within the company, avoiding cross-system switches and manual function searches, boosting efficiency by over 90%.
- Built DQN and PPO models using PyTorch and applied the PPO model to a personal project.
- Acquired proficiency in SSH remote connections, setting up and managing a personal server.
- Gained experience with Docker and Kubernetes, creating and deploying Docker images to the server.

University of Manchester Autumn Game Jam 2024
Oct. 27, 2024 - Oct. 30, 2024

Project: Stop Ghosting Me - github.com/Caesar723/Stop-Ghosting-Me — Manchester, UK

- Developed a game using Unity as the game engine in a two-person team.
- Achieved second place in the competition with innovative multi-window interactions.
- Use of Netcode for establishing local connections and enabling multi-window interactions.

PROJECTS

Magic Fan Made - Personal Project
Jan. 2024 - Present

Game github (includes demo video in the readme): github.com/Caesar723/Magic

Magic Game website: www.xuanpei-chen.top:8000/ (Please use chrome browser to access)

- Built a nearly complete game system with features including quests, shop, card drawing, deck building, creative workshop, roguelike tower climbing, PvP, and PvE modes; the project totals over 32,000 lines of code.
- Created a server image using Docker and deployed it on Alibaba Cloud.
- Made a developer environment image with Docker and Kasm for easy development.
- Developed backend using Python and FastAPI with MySQL and MongoDB for database operations and implemented async features, card drawing, and stack mechanisms.
- Built the frontend using JavaScript and Canvas 2D, enabling real-time interaction via WebSocket.
- Created pseudo-3D effects and smooth animations using matrix transformations and action queues.
- Integrated PPO reinforcement learning as the AI for card gameplay, enabling it to play cards like a human player.
- After the AI Programmer Internship, improved the AI in terms of data encoding and model architecture; achieved over 90% win rate against previous AI versions.
- Enabled users to create their own DIY cards using RestrictedPython for a creative workshop.

Birthday gift for KaKa - Personal Project
Jul. 2023 - Aug. 2023

Github: github.com/Caesar723/Birthday_gift_for_KaKa

- Implemented various fireworks shapes and trajectories using OpenGL, utilizing mathematical and mechanical calculations to simulate realistic motion.
- Designed an efficient particle system using Numpy and memory pool management for optimal resource allocation and utilization.
- Developed a pointer polling algorithm to dynamically allocate particle memory, reducing fragmentation and improving performance.
- Optimized rendering performance with VBO and Display Lists, significantly enhancing the rendering of dynamic and static objects.

TheDayOfSagittarius3 - Personal Project
Apr. 2022 - Jun. 2022

Github: github.com/Caesar723/TheDayOfSagittarius3

- Developed the game core using Python and Pygame, leveraging multiprocessing to separate game logic and network communication for improved stability and responsiveness.
- Implemented asyncio for asynchronous socket communication, ensuring real-time data transmission.
- Connected C++ dynamic libraries using ctypes and numpy.ctypeslib for intensive computations, achieving a 20x perf boost.
- Bypassed Python's GIL using ctypes and multithreading to further enhance performance.

SKILLS

Language: Python, JavaScript, CSS, HTML5 Canvas, C#

Technology stack: Async, FastAPI, Multiprocessing, Multithreading, NumPy, OpenGL, Pygame, PyTorch, Reinforcement Learning, Unsupervised learning, Supervised learning, SQLAlchemy, WebSocket,Pandas

Tools: Docker, Git, MySQL, Vim, MongoDB, Unity